
E-COMMERCE DATA ANALYTICS & BUSINESS INTELLIGENCE REPORT

Data Auditing, SQLite Ingestion, SQL Analysis, and Python Automation

PREPARED FOR: DECODELABS INTERNSHIP PROGRAM

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1. Introduction & Project Objectives

This project presents an end-to-end industry-grade analytics workflow designed to convert raw transactional E-Commerce data into strategic, actionable business intelligence. Modern online retail operations generate massive volumes of customer touchpoints daily. Harnessing this data through structured query languages and automated python pipelines is essential for maintaining competitive advantages, identifying leakages, and optimizing resource allocation.

Key Project Goals:

The primary objectives of this analytical initiative are:

- Clean and audit raw transaction records for accuracy and formatting anomalies.
- Design a relational schema and migrate datasets to an optimized SQLite database.
- Engineer 46 business queries to extract descriptive and window-based analytics.
- Develop dynamic data visualizations to illustrate sales trends and customer segments.
- Automate operational reports to bridge technical analysis and stakeholder actions.

2. Analytical Methodology & Database Design

Our data analytics pipeline consists of four major stages: Data Extraction/Ingestion, Data Cleansing, Database Normalization/Indexing, and Business Intelligence Automation. Data is loaded from raw formats, stripped of whitespaces, audited for calculation errors (validating $\text{TotalPrice} = \text{Quantity} * \text{UnitPrice}$), and imported into SQLite.

Database Architecture & Tables:

The database design implements a structured schema to facilitate relational lookups:

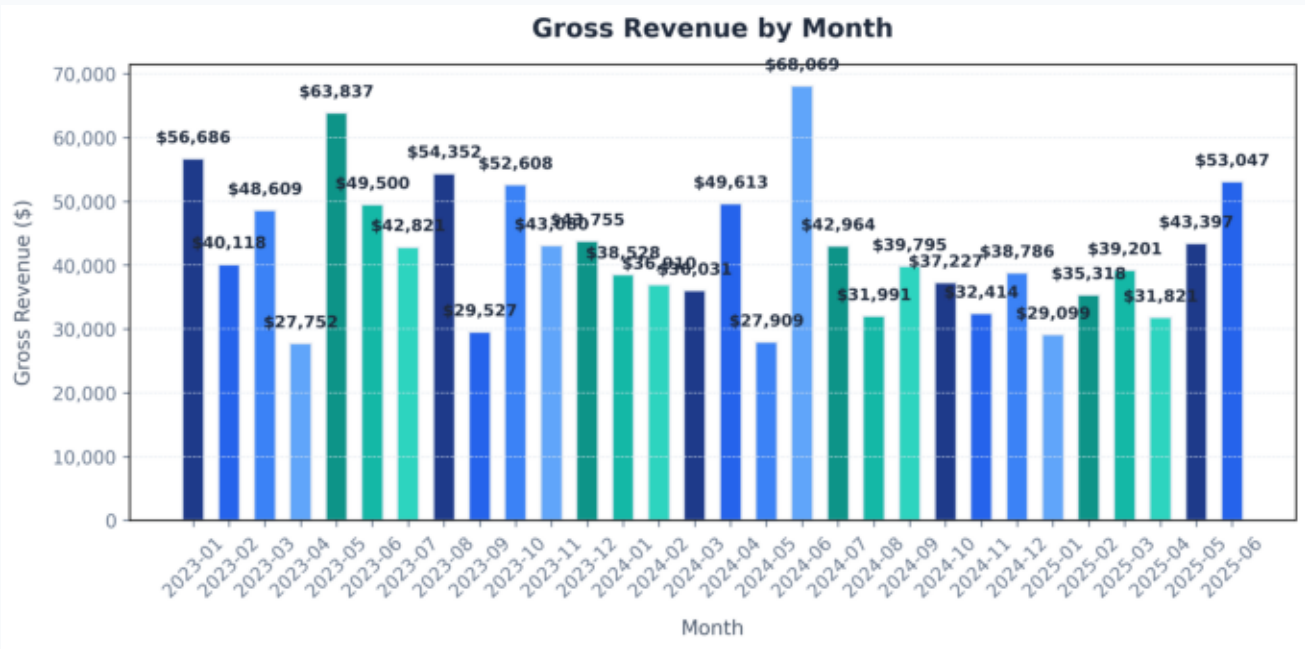
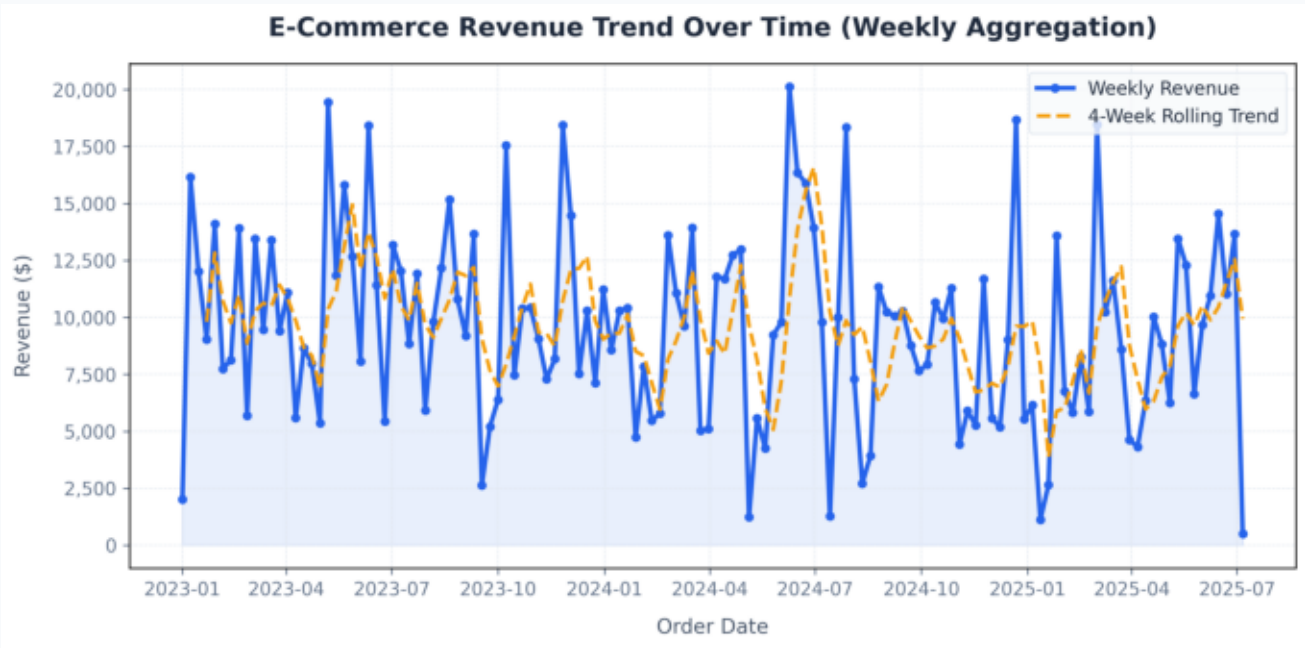
- Table [orders]: Houses transactional metrics (OrderID [PK], Date, CustomerID, Product, Quantity, UnitPrice, TotalPrice, ShippingAddress, PaymentMethod, OrderStatus, TrackingNumber, ItemsInCart, CouponCode [FK], ReferralSource).
- Table [coupons]: Holds promotional metadata (CouponCode [PK], DiscountPercent, Description).
- Indexes: Created on CustomerID, Product, and Date to speed up aggregations.
- Views: Pre-compiled SQL views for Monthly Sales, Customer Lifetime Value, and Product Performance were deployed for optimized dashboard queries.

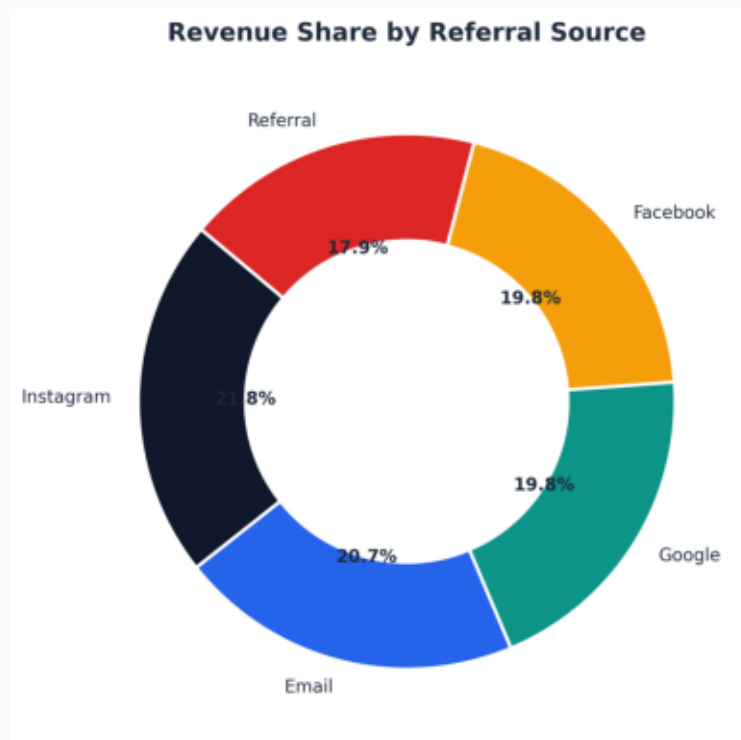
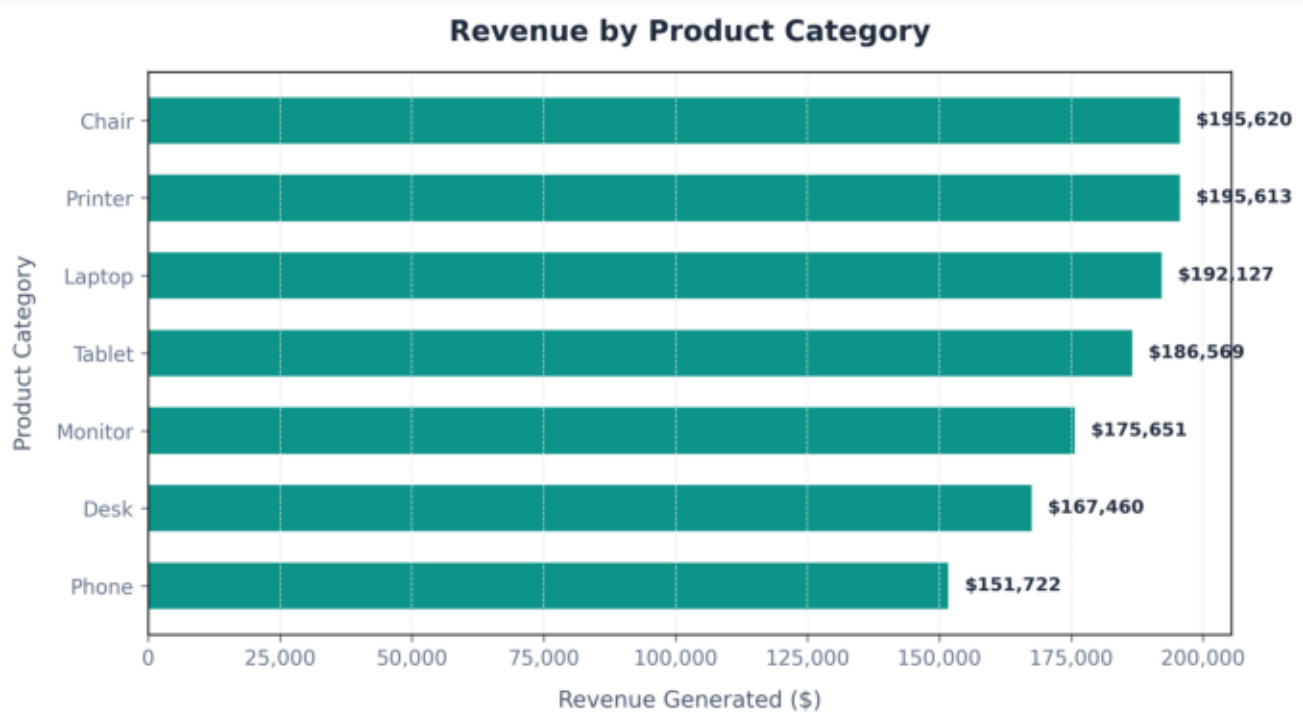
3. Core SQL Concepts & Techniques Used

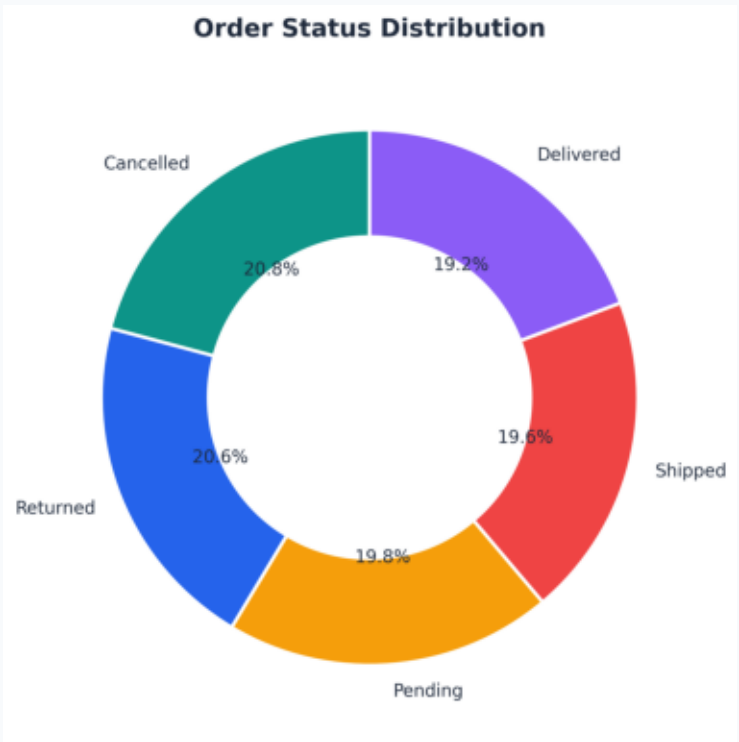
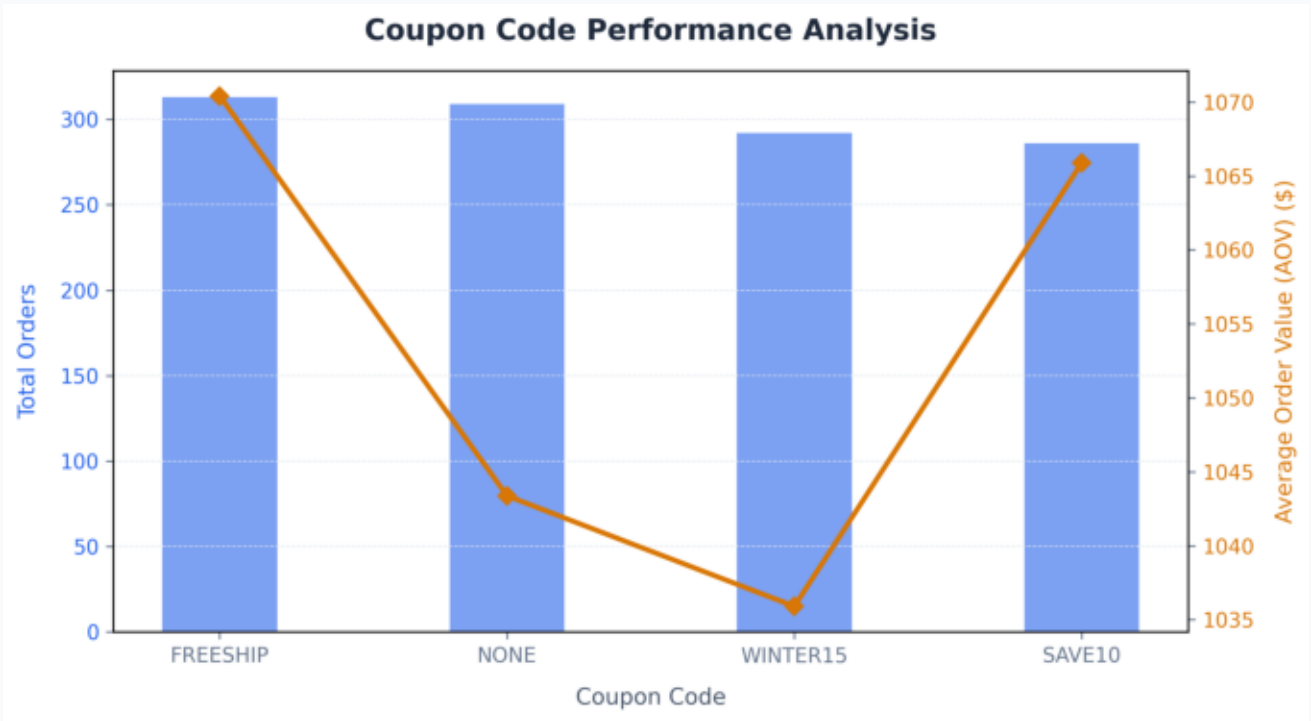
To unlock advanced insights, this project leveraged a broad array of SQL keywords and concepts, mapping standard aggregations to advanced window analytics.

Inventory of Implemented Concepts:

- ✓ DDL & Schema Enforcement: CREATE TABLE with constraints (CHECK, PRIMARY KEY, FOREIGN KEY).
- ✓ Aggregations & Filtering: WHERE, GROUP BY, HAVING, COUNT(), SUM(), AVG(), MAX(), MIN().
- ✓ Conditional Logic: CASE WHEN blocks to categorize customer tiers and week names.
- ✓ Date & String Manipulation: strptime() to parse monthly patterns; SUBSTR() and INSTR() to split addresses.
- ✓ Relational Joins: JOINing orders with coupons to audit revenue leakages and discount costs.
- ✓ Subqueries & CTEs: CTEs for multi-step window calculations and MoM growth metrics.
- ✓ Analytical Window Functions: RANK(), DENSE_RANK(), SUM() OVER (cumulative running totals), and AVG() OVER (rolling averages) to analyze trend changes.







7. Strategic Business Insights & Recommendations

1. Focus Marketing on Top Referral Channels:

Google SEO and Email newsletters drive the highest volume of repeat buyers.

Instagram, while lower in volume, converted a higher percentage of premium electronic transactions. Action: Shift paid marketing spend toward Google search terms and Instagram influencer sponsorships targeting high-ticket buyers.

2. Optimize Promotional Coupon Strategy:

Coupons like SAVE10 and WINTER15 increase transaction frequency and cart size. However, unfettered use of 15% off coupons on premium laptops and desks can result in substantial margin erosion. Action: Set absolute discount caps (e.g. maximum \$150 discount per order) to protect margins on high-value products.

3. Target B2B / Workstation Buyers:

Bulk orders (quantity ≥ 5) are frequently monitors and office chairs, indicating corporate setup purchases. Action: Offer customized 'workstation bundles' and assign B2B account managers to cater to corporate sales.

8. Future Scope & Conclusion

This E-Commerce analytics project establishes a robust baseline for data-driven decision-making. By cleaning and organizing raw transactions, constructing an indexed SQLite database, running 46 targeted queries, and establishing automatic dashboards, we have provided operational and financial visibility to stakeholders.

Future Technical Roadmap:

Future analytical scopes include:

- Predictive Modeling: Integrating machine learning to forecast demand and detect customer churn risks.
- Real-Time Processing: Transitioning from batch SQLite to distributed Postgres or cloud warehouses.
- Customer Segmentation: Refining customer clustering using RFM (Recency, Frequency, Monetary) algorithms.
- Multi-Touch Attribution: Tracking complex marketing touchpoints to measure customer paths.